

Visualizing subnational democracy using Varieties of Democracy (V-Dem) survey data

An application using the case of Colombia

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Abstract

The objective of this methodological paper is to introduce a procedure for operationalizing Varieties of Democracy (V-Dem) data for use at the subnational level. This effort in democracy studies is data-driven and descriptive, and is intended to stimulate discussion on the methodological potential for V-Dem to be applicable within countries. After describing the structure and data generating process behind V-Dem coding, this paper outlines the systematic conceptualization of the background concepts contained within elections and civil liberties surveys, and assigns measures to those concepts on a geographic scale. Once systematized, the data are reduced using a factor analysis technique that calculates latent attributes whose weights can be applied to V-Dem proportional scores. We demonstrate this process for a portion of the 21 response items in the survey, using 2018 cross-sectional data from Colombia at the municipal-level.

1 Introduction

There is a need for subnational data on many political phenomena: violence, democracy, service provision, human rights, gender relations, discrimination, and a range of other topics. The need is especially great in large countries, which are likely to contain considerable geographic variation on these dimensions within their borders. To the extent that such variation exists, claims about the “typical” levels of these dimensions

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for whole countries are less useful, either as descriptions or as independent or dependent variables. In this paper, we demonstrate a technique for combining several underexplored V-Dem variables with geocoded demographic and political data to measure levels of democracy in Colombia at the municipal level.

While data based on direct observation are generally assumed to be superior, there are some drawbacks to this data collection strategy below the national-level. In the first place, hard information about subnational units, such as states or provinces, counties or municipalities, or especially smaller districts, is often scarce. Second, even if the information exists, it can be difficult or expensive to obtain, and may be collected in unstandardized ways. Third, such “objective” data – counts, texts, etc. – can be invalid when they capture only a small part of the concept one needs to measure. It may be feasible to count the number of newspapers and broadcasters in a locality, but a valid measure of freedom of information, for example, should also consider the size of their audiences, their ability to carry out independent investigations, or their editorial autonomy. Fourth, whether a directly observable measure matters is often conditional on other factors. The lack of arrests of journalists in North Korea is not evidence of press freedom; governing parties sometimes win elections for decades even when elections are very competitive; party platforms and speeches are often insincere and inaccurate; and texts of constitutions and laws can be “parchment” rules, dead letters that are not implemented in practice.

For these reasons, there are sometimes advantages to following an alternative measurement strategy: relying on the subjective judgments of multiple country experts. This approach can compensate for some of the weaknesses of “objective” data (Bernhard et al. 2017). Experts absorb information from multiple sources: reading, following news, conversations, personal experiences, and deep familiarity with the social and political context. This enables them to reconcile diverse and often conflicting kinds of information and synthesize it into an internally consistent account of how things work. Their understanding extends to a sense of the past and the direction of trends. Because they possess this depth of understanding, experts are less likely to be deceived by insincere statements or *de jure* arrangements that are ignored in practice, and they are less likely to consider one attribute definitive when they know that other conditions render it irrelevant.

Of course, there is also an obvious disadvantage to subjective knowledge: different experts tend to synthesize somewhat different understandings. They arrive at different interpretations of the same reality. This is why it is crucial to consult multiple experts, which makes it possible to have greater confidence in the interpretations on which experts agree, and to exercise prudent caution about matters on which there is little agreement. Post hoc validation is also essential. There are no guarantees that subjective ratings will yield useful information, but it is worth trying as long as we also subject the outcomes to multiple forms of validation.

1.1 Measuring subnational democracy

Subnational democracy is an area where subjective expert ratings have the potential to improve measurement. There are hundreds of democracy measures, but almost all of them are at the national level of analysis (Bernhard, Nordstrom, and Reenock 2001; Bertelsmann Stiftung 2005; Boix, Miller, and Rosato 2013; Bühlmann, Merkel, and Wessels 2008; Coppedge et al. 2023; *Democracy Index 2010: Democracy in Retreat* 2010; House 2015; Marshall and Jaggers 2014; Pemstein, Meserve, and Melton 2010; Przeworski and Limongi 1997; Skaaning, Gerring, and Bartusevičius 2015; Vanhanen 2000). The few exceptions (Gervasoni 2010; Giraudy 2013; McMann and Maguire 2018; Mickey 2015) cover only one country (although Giraudy does two).

In a recent methodological effort, Pérez Sandoval (2023) designed an Index of Subnational Electoral Democracy (ISED). The ISED tracks electoral outcomes at the provincial level (the first jurisdictions below the national-level) for 11 countries over 40 years, using a framework attuned to the complexities around territorial conflation around country-level systems such as judicial branches and political parties. However, this index relies on electoral data rather than country expertise. As alerted to earlier, electoral data may be collected in non-standardized ways, especially in local jurisdictions. More importantly, the number of parties may not always signify a competitive political climate, nor does vote count always imply effective democratic participation. Without considering the relevance of other factors, such as the strength of civil liberties, electoral information alone will not always be an efficient or valid measure of subnational democracy.

In addition to providing a novel subnational index, Perez Sandoval provides a helpful summary of the concepts, operations, and data collection methods used by other researchers of subnational democracy. Among the eleven authors cited, only four – Hernández Valdez (2000), Ortega and Somuano in Meixueiro Nájera and Moreno Pérez (2014), Gervasoni (2018), and Grumbach (2023) – attempt to measure beyond the electoral dimension of democracy at the subnational level. These authors include measures of civil liberties components – such as human rights violations, local radio and newspapers, or institutional constraints – however they do so only for individual countries (Mexico, Argentina, and the United States) and at the provincial or state level. All four authors use advanced data reduction techniques, such as Bayesian factor analysis or principal component analysis, to arrive at subnational democracy scores, however they stop short of providing a method for cross-national comparison.

Finally, pivoting to subjective expert data, McMann et al. (2021) use a selection of V-Dem variables to measure “unevenness” of elections and civil liberties in most countries, but theirs is a national-level concept that tells us nothing about where democracy is better or worse within a state.

1.2 V-Dem subnational data

V-Dem data contains several variables that provide a partial path to measurement at the subnational level. There are four questions on elections and civil liberties: Subnational election area less free and fair characteristics (`v2elsnlfc`), Subnational election area more free and fair characteristics (`v2elsnmrfc`), Stronger civil liberties characteristics (`v2clrgstch`), and Weaker civil liberties characteristics (`v2clrgwkch`). Although elections and civil liberties are far from capturing every aspect of democracy, they cover the two most central criteria for polyarchy (Dahl 1971).

Table 1: V-Dem questions and clarifications

<i>Question: How would you describe the areas of the country in which elections are significantly less free and fair? [v2elsnlfc]</i>
<i>Question: How would you describe the areas of the country in which elections are significantly more free and fair? [v2elsnmrfc]</i>
<i>Question: How would you describe the areas of the country where government officials' respect for civil liberties is significantly stronger? [v2clrgstch]</i>
<i>Question: How would you describe the areas of the country where government officials' respect for civil liberties is significantly weaker? [v2clrgwkch]</i>
Clarification: Choose all that apply.
Responses:
0: Rural. (0=No, 1=Yes)
1: Urban. (0=No, 1=Yes)
2: Areas that are less economically developed. (0=No, 1=Yes)
3: Areas that are more economically developed. (0=No, 1=Yes)
4: Inside the capital city. (0=No, 1=Yes)
5: Outside the capital city. (0=No, 1=Yes)
6: North. (0=No, 1=Yes)
7: South. (0=No, 1=Yes)
8: West. (0=No, 1=Yes)
9: East. (0=No, 1=Yes)
10: Areas of civil unrest (including areas where insurgent groups are active). (0=No, 1=Yes)
11: Areas where illicit activity is widespread. (0=No, 1=Yes)
12: Areas that are very sparsely populated. (0=No, 1=Yes)
13: Areas that are remote (difficult to reach by available transportation, for example). (0=No, 1=Yes)
14: Areas where there are indigenous populations. (0=No, 1=Yes)
15: Areas where the national ruling party or group is strong. (0=No, 1=Yes)
16: Areas where the national ruling party or group is weak. (0=No, 1=Yes)
17: Areas that were subject to a longer period of foreign rule. (0=No, 1=Yes)
18: Areas that were subject to a shorter period of foreign rule. (0=No, 1=Yes)
19: Areas that were recently subject to foreign rule. (0=No, 1=Yes)
20: Areas that have not recently been subject to foreign rule. (0=No, 1=Yes)
21: None of the above. (0=No, 1=Yes)

Table 1 reproduces the four questions, along with the same 21 “choose all that apply” criteria that apply to all questions. In the country-year V-Dem data sets, each criteria becomes a separate variable that is no longer binary, but rather the proportion of experts who chose the “yes” response to that option. Thus, in

actuality we have 84 (21*4) variables to work with.

These unusual options have advantages and disadvantages. On the positive side, these options ask experts to identify the criteria that distinguish locations where democracy is better from locations where it is worse. The options are described in a generic way so that they could be relevant in any country that V-Dem rates, anywhere in the world, and over more than two centuries of history. In this paper we deal with 16 of them, leaving aside the options referring to foreign rule (# 17-20) which is not very relevant for Colombia, and option #21 which refers to a negative response (“None of the above”). Still, even these few response options are varied enough to cast a wide net that captures diverse criteria for distinguishing where democracy is better or worse.

On the negative side, first, all of these options are conceptually binary. In effect, they ask “Is this a useful discriminating criterion or not?” rather than seeking information about degrees of relevance. Second, the options do not name a specific concept or indicator that should be used in empirical analysis. This decision is left up to each country expert and V-Dem does not reveal—and in fact does not *know*—on what information country experts based their assessments. Third, the options do not specify a threshold that should be used to decide whether a location does or does not satisfy the criterion. Fourth, many of the criteria overlap and are therefore somewhat interchangeable. This is especially likely for the binaries of rural-urban, low-high income, sparsely-densely populated and remote areas, which may also overlap with areas of illicit activity, civil unrest, and indigenous population. Finally, the criteria are not mapped onto well-defined locations such as states, counties, or cities that would have known administrative boundaries.

Given these features, additional work is required to render the V-Dem data useful for measuring democracy at the subnational level. The procedure described below is structured around the following steps: coder-level analysis, selection of indicators (operationalization), data reduction, and finally mapping.

2 Coder-level Analysis

The degree of agreement among experts can be interpreted as a graded measure of the degree to which each criterion is likely to be useful for discriminating between better and worse locations. Arguably, it is also reasonable to interpret these proportions as, in part, reflections of degrees of fair elections or civil liberties because the more democratic a subnational unit is, the more likely experts are to notice it. In effect, we assume that experts perceive degrees of democracy, but also have their own subjective thresholds on that continuum that they use to distinguish more democratic situations from less democratic ones when they are asked to make a binary classification. Experts’ thresholds differ, but the more democratic a unit is on

Table 2: V-Dem response scores for Colombia in 2018

Response Criteria	Elec. less free	Elec. more free	Civil lib. stronger	Civil lib. weaker
Rural	0.875	0.000	0.000	1.0
Urban	0.125	1.000	0.889	0.0
Less econ devt	0.875	0.111	0.000	1.0
More econ devt	0.125	0.667	1.000	0.0
Inside capital	0.000	0.889	1.000	0.0
Outside capital	0.375	0.000	0.000	0.6
North	0.250	0.222	0.333	0.1
South	0.250	0.000	0.000	0.5
West	0.125	0.000	0.111	0.3
East	0.125	0.000	0.111	0.3
Civil unrest	1.000	0.000	0.000	0.9
Illicit activity	0.875	0.000	0.000	0.9
Sparse pop.	0.375	0.000	0.000	0.6
Remote	0.875	0.000	0.000	0.8
Indigenous	0.500	0.000	0.000	0.9
Ruling party strong	0.000	0.222	0.333	0.0
Ruling party weak	0.000	0.000	0.000	0.1

the underlying continuous dimension, the larger the proportion of experts that will judge it to be in the “more democratic” category. This psychological model is the basis for the Bayesian Item Response Thoery (IRT) measurement model that V-Dem uses to aggregate multiple coders’ ratings into point measures for country-years (Marquardt and Pemstein 2018; Pemstein et al. 2023). Although it would be ideal to utilize a comparable model for this project, we lack sufficient data to attain the statistical power equivalent to that available in V-Dem. Instead, we rely on coder-level information to assess the relative importance of the subnational survey criteria for a given country in a given year.

2.1 Coder proportion scores

Coder scores for each criteria-year observation are proportions. However, these proportional scores are “noisy” data, in the sense that coders may not have access to country-wide municipal data on phenomena such as illicit activity or population sparsity. Moreover, some information may be missing especially in areas with weak institutional capacity.

To visualize how these scores vary across survey items, Table 2 shows the proportional scores for V-Dem data subset to Colombia for 2018. We use these scores to demonstrate the mapping procedure below. As the table illustrates, some survey criteria register high degrees of variation across surveys (e.g., Rural or Urban, Economic development, Capital city, Civil unrest, Illicit activity) while others register relatively

lower variation (e.g., North-South-West-East, Ruling party strong or weak). However, in this case, extreme ends of the range (scores closer to 0 or closer to 1) indicate more agreement around the salience of a criteria, and conversely, scores towards the middle (closer to 0.5) indicate lower amounts of consensus among voters.

With regard to scores of 0, the data do not distinguish between coders who selected “no” and coders who did not respond. For example, for the survey question on “Elections that were less free and fair” in 2018, a total of eight coders responded. However, we are unable to determine (given certain use restrictions around the data) whether the final score 0.000 for “Areas where the ruling party is weak” results from a particular combination of coders, say five that agreed “no” this is not a relevant criterion, and another three did not respond (“NA”) because they did not have an opinion. For example, in the case of Indigenous, the final score is 0.500 but all we know is that half of all coders thought this was relevant and the other half thought it was not or had no opinion.

2.2 Confidence scores

In addition to the final proportion of responses, V-Dem also collects the level of self-reported confidence at the individual coder-level, for each survey, scaled from 0 to 100. In theory, this could provide a useful metric by which to condition or weight responses. For example, the average level of confidence for coder-level responses to “More free elections” criteria in 2018 was 76.43 compared to 90.00 for “Less free elections.” These values could be applied to the final scores for each category at the municipal-level resulting in a differential estimation of the characteristics affecting subnational elections. However, this method would add another layer of complexity because self-reported confidence is a subjective and non-standard practice. Individual coder reliability scores exist, and they are used for calculating point scores for the final indices in V-Dem, however these are unpublished. (Some coder-level information are available, and we could request these details if necessary.)

2.3 Numbers of coders

A high number of coders may shy away from even providing responses to the four subnational democracy questions in the V-Dem surveys. There are 21 questions in the elections survey, of which only 2 are subnational, and 24 questions in the civil liberties survey, of which only 2 are subnational. Figure 1 shows the difference in the average number of coders (V-Dem variables ending in `*_nr`) who responded to each category of questions for the topics of elections and civil liberties in Colombia from 1900-2022. Subtracting the average number of coders that responded to subnational questions from the average number of coders

that responded to non-subnational questions, the differences have always remained positive. Fewer coders respond to subnational questions than to non-subnational questions. Because fewer coders decide to respond to these questions, this may suggest a lack of conclusive information or familiarity around the topic of subnational democracy in Colombia, at least as conceptualized using V-Dem criteria. Comparing across surveys, the elections questions register a higher average difference compared to civil liberties questions, escalating notably after the direct elections of mayors was introduced in 1988. There is a major escalation in elections-related differences to about 8.5 coders between 2005-2012, while the difference in civil liberties-related questions has declined steadily over time. This jump may have been a result of V-Dem bridge coders incorporated in a recent version of the survey.

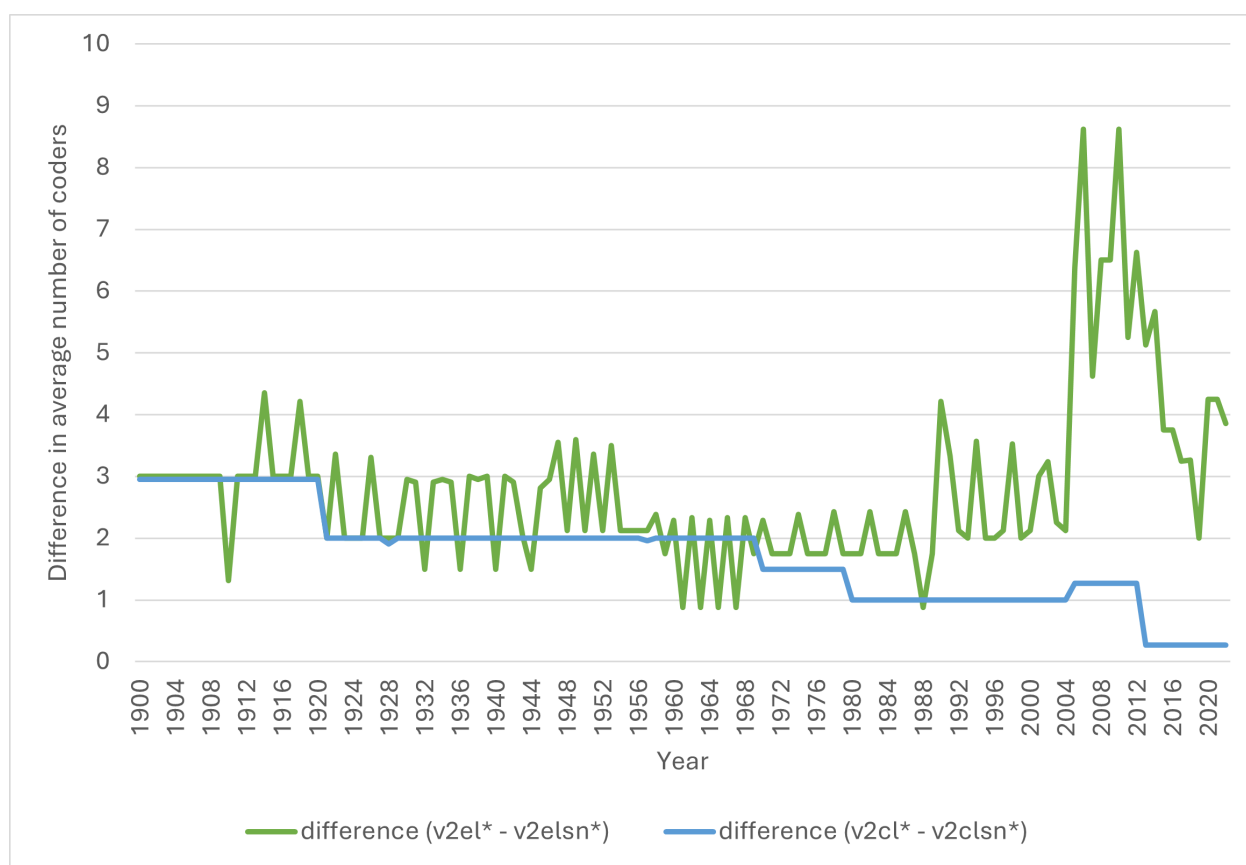


Figure 1: Difference between the average number of coders that responded to the national-level survey questions and the average that responded to subnational-level questions, in the case of Colombia.

These differences in the overall number of coders and the number of coders who responded to subnational questions provide a relevant sense of the “attrition” rate of coders. This information can be useful for assessing the relative reliability of the proportional scores for a given year. For example, in Colombia for 2018, the number of respondents for each subnational survey question and maximum number for all other survey questions in the survey were as follows:

- Elections less free and fair: 8 over 13 maximum coders
- Elections more free and fair: 9 over 13 maximum coders
- Civil liberties stronger: 9 over 10 maximum coders
- Civil liberties weaker: 10 over 10 maximum coders

These attrition rates suggest that civil liberties data have more inter-coder reliability than election data. When operationalizing the criteria from each survey, these coder attrition rates can serve to temper the interpretation of composite scores for subnational democracy. In this case the relative reliability of proportional scores for “Elections less free and fair” will be lower than those for “Civil liberties weaker.” This means that resulting estimates for the civil liberties dimension of subnational democracy in Colombian municipalities in 2018 will be of higher confidence than those for elections.

Additional analysis of coder-level behavior may inform the weighting procedure further. For example, bridge coders recruited in 2000s may explain the escalation in differences between coders in national versus subnational-level questions. Bridge coders typically code more than one country, but may be hesitant to respond to subnational-level questions, particularly if they reside outside of the country under analysis. Controlling for bridge coders would lead to different weighted estimates of subnational democracy. Another exercise could include comparing individual coder behavior by year: some individuals may be more prone to attrition during a certain period, such as 2005-2012, compared to those just in that period.

3 Selection of indicators and measurement

In order to operationalize V-Dem criteria for a subnational context, we choose one or two specific indicators – from official state data or others’ scholarly research – to measure them. For each indicator, we either choose a threshold to represent the binary concept of whether the criterion applies to a location or not; or we redefine the concept to be continuous so that it can measure the degree to which each location satisfies the criterion. In this case, most indicators work well as matters of degree: income, population density, strength of the ruling party, etc. We prefer the continuous criteria where possible. However, other criteria are probably best left binary, such as the categories of North, South, West, and East – although even these could be measured as fuzzy sets.

The country context is important for defining concepts and selecting appropriate indicators. This occurs for both fuzzy and less fuzzy concepts. For example, civil unrest and illicit activity are generally contested or

Table 3: Operationalization of V-Dem responses in Colombia

ID	Response	Indicator
0	Rural	Rural Index (DANE)
1	Urban	
2	Less econ devt	Mun. Value Added / IPM (DNP)
3	More econ devt	
4	Inside capital	Geographic distance from center (DANE)
5	Outside capital	
6	North	Six macro-regions (DANE)
7	South	
8	West	
9	East	
10	Civil unrest	Forced displacement / Confinements (Min. Defense)
11	Illicit activity	Robberies / Homicides / Coca cultivation (Min. Defense/UNODC)
12	Sparse pop.	Population density (DANE)
13	Remote	Distance to market (CEDE))
14	Indigenous	Ethnic populations (DANE)
15	Ruling party strong	Pres. margin of victory (RNEC)
16	Ruling party weak	

fuzzy concepts. Moreover, they are difficult to observe and isolate as factors that affect democracy in the same way across contexts, even within countries. Taking illicit activity, for example, one area of the country with gang controlled neighborhoods may have limited voter turnout and less free elections, while another may tolerate or even encourage electoral competition (Córdova 2022). Even basic categorizations may be contested. For one expert, the North may include 10 provinces, yet for another it may include 8. Some knowledge about these context-specific contested concepts should inform the conceptualization process from the start, as these decisions will ultimately shape how subnational democracy levels are mapped.

In the following subsections, we justify our choices of indicators for the discriminating criteria and explain how we convert the majority to continuous scales using the cumulative distribution function (CDF). The CDF standardizes each criterion on a 0-1 scale so that they are more easily comparable.

Additionally, we combine natural sets of criteria that can be operationalized by the same indicators. For example, the concepts of rural and urban can be measured by a rurality index, and less or more economic development can be measured by a continuous scale of GDP per capita or multidimensional poverty index. This has implications for our measurement approach using V-Dem proportional scores, because these too must be combined correspondingly. In Colombia, we consider the indicators as shown in Table 3. All sources in parentheses are official government sources, although alternative indicators and sources certainly exist.

3.1 Criteria #0-1: Rurality

The survey asks about elections and civil liberties in rural or urban locations. Rurality or urbanization can be measured in many ways. The World Bank’s Rural Access Index (RAI)¹ or European Commission’s Global Human Settlement Layer (GHSL)² are exemplary GIS-based approaches. (Waldorf and Kim n.d.) create an index that considers population size, density, remoteness, and built-up areas. In Colombia, the National Statistics Administrative Department (DANE) uses a classification based on the Organization for Economic Co-operation and Development (OECD) criteria that defines rural territories as those with 150 or fewer persons per kilometer squared (km^2). Using this criteria, the National Planning Department (DNP) finds that over 75% of Colombian municipalities do not meet the threshold of even 100 persons per km^2 . From another perspective, DNP estimates that close to 4/5 of the population lives in urban centers. (Departamento Nacional de Planeación 2014)

For operationalizing the rurality criterion, we use a demographic measure that considers the proportion of the municipal rural population over the total municipal population. This basic “Rurality Index” measure has a few advantages. First, it was created by the Economics Center for Development and Economy (CEDE) at *Universidad de Los Andes*, using official Census data from DANE. Second, this proportional measure provides a continuous scale of municipal rurality that can facilitate cross-municipal comparison.

3.2 Criteria #2-3: Economic development

Economic development appears in the subnational V-Dem criteria under responses 2 (“Areas that are less economically developed”) and 3 (“Areas that are more economically developed”). A common measure of economic development is GDP per capita, however this measure is problematic when scaled down to the subnational level. According to DANE, only 41 countries in the world calculate regional or subnational GDP (“Buenas prácticas internacionales PIB regional” 2020). The difficulty may stem from the complex task of isolating economic productivity for subnational units when inhabitants may be undertaking inter-municipal daily commutes, for example, or when supply-chains overlap with multiple departmental jurisdictions. DANE provides subnational data for GDP (or *Producto Bruto Interno*, PBI, in Spanish) at regional and departmental (provincial) levels, but not at the municipal level.³

¹See: <https://datacatalog.worldbank.org/search/dataset/0038250>.

²See: <https://ghsl.jrc.ec.europa.eu/CFS.php>.

³Most departments are covered by the GDP measure, but newer departments (many of which were inducted in 1991) are grouped into one unit, and therefore cannot be mapped based on DANE’s Departmental Code system without data intervention. These are: Amazonas (91), Arauca (81), Casanare (85), Guainía (94), Guaviare (95), Putumayo (86), San Andrés, Providencia y Santa Catalina (Archipiélago) (88), Vaupés (97) y Vichada (99).

At the municipal-level, DANE has created a measure of municipal “value added”, which estimates the economic capacity of municipalities within each department by subtracting consumption levels from gross production (Departamento Administrativo Nacional de Estadística 2022). Nearly all municipalities are measured by the Municipal Value Added (VAM) measure, however, data only exists from 2011 onward. Additionally, DANE estimates a Multidimensional Poverty Index (IPM) consisting of 5 dimensions: education levels, childhood conditions, employment, health, and access to public services and housing conditions (Departamento Administrativo Nacional de Estadística 2021). Using these two municipal-level indicators produces a more comprehensive operationalization of more or less economically developed areas.

3.3 Criteria #4-5: Inside or outside of capital city

The surveys ask how elections or civil liberties vary “Inside / Outside the capital city.” In the context of the V-Dem survey, which is cross-national, respondents may reasonably interpret this question to refer to the national capital (Bogotá). *University of Los Andes* Center for Studies on Economic Development (CEDE) provides a measure for this variable (linear distance in kilometers). Although the survey asks two questions, the first (“Inside the capital city”) is one extreme of the second (“Outside”), so we create one continuous variable that reflects the relative proximity of municipalities from Bogotá.

3.4 Criteria #6-9: North, South, West, East

Although most criteria in the subnational V-Dem questions can be conceptualized using continuous measures, some criteria are less amenable to this operation. A clear example are criteria #6-9 which ask respondents to assess elections or civil liberties across four cardinal directions: North, South, West, and East. While these categories could be measured using fuzzy sets, where middling municipalities would have multiple memberships to different regions, we use a binary measure of membership, grouping together pre-existing regional categories. Rather than using four cardinal regions in Colombia, DANE classifies the country into six “macro-regions”: a) Caribe, b) Eje Cafetero (Antioquia), c) Centro Oriente, d) Llano, e) Sur, and f) Pacifico (“DANE - información del DANE para la toma de decisiones en departamentos y ciudades capitales” n.d.).⁴ We group the six macro-regions in the following manner: a)-b) as “North”; c)-d) as “East”; e) as “South”; and f) as “West”.

⁴A public-access table of municipalities classified into these six regions can be found [here](https://www.datos.gov.co/Mapas-Nacionales/Departamentos-y-municipios-de-Colombia/xdk5-pm3f/about_data). There are other divisions, such as the five regions used by CEDE studies (Andina, Caribe, Pacifica, Orinoquia, Amazonia), but for now we go with the prior classification.

3.5 Criterion #10: Civil unrest

The survey questions ask to identify whether elections or civil liberties are stronger or weaker in “Areas of civil unrest (including areas where insurgent groups are active).” This criterion can be operationalized using many potential indicators, including the presence of insurgent forces, events of forced confinement, protests and mobilizations, or forced displacement, among others. For the purposes of this paper, we use reported cases of forced displacement and forced confinement events. Both indicators are measured by Colombia’s national Victims’ Registry, in place since 2005.

3.6 Criterion #11: Illicit activity

The survey questions also ask to identify the strength of elections or civil liberties in “Areas where illicit activity is widespread.” This concept can take many forms, especially given characteristics of the local environment. We consider three indicators of widespread illicit activity: robberies, homicides, and manual illicit crop eradication. The data for the first two indicators are compiled by the Ministry of Defense, and coca eradication data are compiled by Colombia’s Drugs Observatory.

3.7 Criterion #12: Sparse population density

The surveys ask coders whether stronger or weaker elections or civil liberties are correlated with “Areas that are very sparsely populated.” As mentioned before, population density is used to measure levels of rurality. In this case, however, we include a raw population density measure of total inhabitants per km^2 rather than differentiate between urban and rural populations. These data come from DNP, using information from DANE and the Geographic Institute Agustin Codazzi (IGAC).

3.8 Criterion #13: Remoteness

For the criterion “Areas that are remote (difficult to reach by available transportation, for example),” we use a measure of the linear distance of a municipality to the nearest supermarket (*mercado mayorista*), as calculated by CEDE. If a supermarket is located in the municipality, this takes a value of 0. Alternative measures may include distance to the nearest secondary or tertiary road or to a major city.

3.9 Criterion #14: Indigenous

The surveys ask about elections or civil liberties in “Areas where there are indigenous populations.” This is salient in Colombia, where the 2018 Census identified indigenous to comprise 13.6% of the total population (Departamento Administrativo Nacional de Estadística 2019). We include this demographic information, and also include the population of inhabitants that identify as pertaining to an ethnic group. Both of these measures are based on the 2018 Census data from DANE.

3.10 Criteria #15-16: Support for national ruling party

The survey questions ask whether elections or civil liberties vary according to “Areas where national ruling party or group is strong / weak”. We utilize official voting data comes from the National Civil Registry (RNEC) to calculate the percentage of votes cast for the president-elect vs. the runner-up, in 2018. In 2018, there was a runoff election (second round) between Gustavo Petro and Ivan Duque; Duque won, so his is the ruling party. This makes things easier because there are only 2 candidates, the winner and runner-up. Elections without second rounds (prior to 1994, as well as 2002 and 2006) would require additional coding. To operationalize this criteria, we create a continuous measure of the percent margin of victory at the municipal level (at the national level, Duque’s margin of victory was close to 13 percentage points in 2018).

4 Data reduction

Overlapping criteria could be an advantage if they are treated as substitutable, and therefore aggregated with an additive or weighted average formula (Goertz 2006). The underlying logic is that strength on one criterion tends to compensate for weakness in another. This kind of relationship can be operationalized as a sum, an average, or a weighted average. Ideally we prefer weighted averages, as averages preserve the 0-1 range of values and weights employ a more relaxed assumption than “unweighted” averages, which actually enforce a strong assumption of equal weighting. Weights can be derived empirically via a data reduction technique such as factor or principal components analysis, which yield factor or component weights based on the commonalities among our criteria. These techniques can also evaluate whether it makes sense to treat all these criteria as unidimensional. If it does not, then we would do well to generate multiple indices that measure somewhat independent dimensions of democracy.

In this case, we use a factor analysis data reduction technique to identify latent factors that emerge from the data. Exploratory factor analysis using the set of variables associated with V-Dem criteria provides a

set of latent factors with different relative weights for each variable.

Loadings:				
	MR3	MR1	MR2	MR4
IndRur_0t1c	-0.522	-0.169	0.474	
VAM18_2t3c	0.890	0.322	-0.491	
IPM18_2t3c	-0.256	0.187	0.779	0.503
DisBog_4t5c	0.137	0.433	0.231	0.844
Desp_10c	0.658	0.435		0.210
Conf_10c			0.140	
Errad_11	0.109		0.120	
Hurto_11c	0.882	0.342	-0.556	
Homic_11c	0.813	0.473	-0.149	0.196
PobTot_12c	0.936	0.524	-0.371	0.350
DenPob_12c	0.399		-0.786	
DisMer_13c		0.284	0.405	0.374
PobInd_14c	0.585	0.967		0.372
PobEtn_14c	0.669	0.904		0.582
PobInd_14cp	0.350	0.951	0.134	0.325
PobEtn_14cp	0.423	0.898	0.112	0.567
RulPar_15t16c	-0.258	-0.520		-0.443
	MR3	MR1	MR2	MR4
SS loadings	5.235	4.990	2.456	2.424
Proportion Var	0.308	0.294	0.144	0.143
Cumulative Var	0.308	0.601	0.746	0.889

Figure 2: Results from the factor analysis

Figure 2 depicts the results of our data reduction technique, revealing 4 latent factors corresponding to demographic and political attributes at the subnational level in Colombia for 2018. These dimensions can be characterized as such:

- MR3 = Richer, populated, urban environments with everyday violence (#2-3, 12, and 11)
- MR1 = Indigenous concentrations with less official party (Duque) supporters (#14 mainly, some of 12, and maybe 15-16)
- MR2 = Poor, rural disperse, and insecure areas (#12, 2-3, 11)
- MR4 = Peripheral and poor ethnically populated areas (#4-5, 14, 2-3)

The total variance explained by these four factors is 88.9%. The results suggest that the first two factors explain 60% of the data variation, and these are characterized by either more economically developed,

populated areas, or less populated but concentrated indigenous and ethnic areas with less crime. Crime appears to group together with both more and less economically developed areas. Poorer areas distant from Bogotá explain a large portion of variation on its own (over 14%).

[pending further analysis, as are the following sections]

5 Weighting V-Dem coder responses

Our next task is to explore ways to combine these factor scores (and future scores based on additional criteria such as civil disturbances, illicit activities, remote areas, and indigenous population) into a summary measure of a concept closer to “democracy.”

6 Conclusion

As noted previously, validation is a crucial step in producing data based on expert ratings. We have not yet been able to do a systematic validation of the data we have so far, but we have some observations about face validity based on the maps we have produced. In general, our data indicate higher levels of democracy (both in free and fair elections and in protections for civil liberties) in the major urban areas and lower levels in rural or remote areas such as the Amazon. [To be fleshed out after we have a summary map.]

Systematic validation will require comparisons with independent assessments of subnational democracy in Colombia. An initial source for such information will be responses to open-ended versions of V-Dem’s four questions provided by a small number of university and master’s students at a Colombian university in February 2024. They were asked to identify the most salient locations (defined any way they chose) in Colombia and years where they would expect valid indicators to show better or worse situations. We can check our data against their expectations. At some point we could solicit the same kinds of responses from a more senior panel of Colombian experts. Ultimately, the best kind of validation would be administering an online survey asking V-Dem’s country experts on Colombia to code these same questions for Colombian departments or municipalities for selected years. We do not know whether such an effort is feasible, but it is probably impossible to know until we try.

Eventually we hope to extend this exercise to other countries where others have generated measures of subnational democracy, such as Argentina, Mexico, India, the United States, and Russia. We do not downplay the difficulty of such an effort. It requires access to very specific geo-coded data on multiple demographic

and political indicators, which are probably not available for many countries. Furthermore, producing cross-nationally comparable data involves an additional layer of complication. The data we have produced for Colombia is relative: the degree to which local situations are better or worse than the national norm. Converting such relative data to absolute levels will require both anchoring the ratings to a national score and establishing the range of variation within which the relative scores can vary. V-Dem’s Clean Elections Index could serve as the anchor for the subnational elections variables and its Civil Liberties Index could serve as the anchor for the subnational civil liberties variables. The same anchoring indices could be used for other countries. However, we do not yet have a proposed solution for setting the range of variation around these anchors. This would require careful conceptual and methodological effort.

Colombia is among the best cases for this initial proof-of-concept research due to the well-developed geographic data that DANE provides (and for which the Lucy Institute, where co-author Matthew Sisk works, has direct access). Replication in other countries would make it possible to reach more general conclusions about the validity of the technique and to decide whether it is promising enough to extend to many more countries. If results are encouraging, there may be a potential for extending of this technique to measure other political concepts subnationally.

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